Skeletal Data based Activity Recognition System

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Abstract—**In recent years, Human activity recognition and research related to it are in high demand owing to its application in various fields such as healthcare systems, assisted living, surveillance etc. Activity recognition system in general aims at identifying activities performed by person in an environment. In this work, skeletal data-based activity recognition system is presented. Microsoft Kinect sensor is a motion capture sensor developed by Microsoft for xbox one. It has become most popular for its effortless operation and low cost. The 3D skeletal joint positions obtained from this kinect sensor are used as raw data for classification purpose. Set of statistical features are computed from these skeletal data. The dimensions of statistical features computed are reduced to eliminate correlated and redundant features among them. Principal component analysis (PCA), a technique used for finding smaller number of uncorrelated data is employed for reducing the feature dimension. The dimensionally reduced features are used as training data for training the classifier. The ability of K-nearest neighbour (KNN) classifier and Support vector machine (SVM) classifier in classifying actions are analysed. The classification method proposed is tested on most popular KARD (Kinect activity recognition dataset) dataset and various classification parameters are computed. Among the two classifiers considered, SVM classifier performed better with an average overall accuracy of 97% which is 3.44% higher than the accuracy produced by KNN classifier.**

Index terms—**Kinect sensor, Activity recognition and PCA.**

I. Introduction

HUMAN Action Recognition (HAR) expects to distinguish the activities completed by an individual given a lot of perceptions and the encompassing condition. Activity recognition system has gained significant importance in recent years due to huge requirement of security and surveillance especially in crowded areas like malls, social meetings etc. Most of the activity recognition system nowadays uses wearable that produce raw signal patterns when a user performs a particular action.

In [1], authors have used IMUs for automatic detection and segmentation of daily life activities. Agarwal et al in [2], presented a human activity recognition system using Depth sensor. Fast Dynamic Time Warping algorithm with weighting functions were used by them for classification.

In [3], authors have used biaxial accelerometer sensors to detect physical activities. They have placed five sensors at different body parts and classified actions based on the signals simultaneously produced from all five sensors. Similarly, authors in [4] have used multimodal wearable sensors placed at different body parts to classify complex, in-home activities and have obtained acceptable classification accuracy.

Chen et al in [5] provided a detailed survey report on use of both vision and inertial sensors for action recognition. They claim that improvement in recognition accuracy was observed upon fusing vision and inertial sensor data. Cippitelli et al [6], developed an elderly monitoring system to monitor aged people in home. Different types of sensors were used by them to address the task.

In [7], authors have used Kinect sensor for action recognition. Three distinct classification methods namely K-means clustering, support vector machine and hidden markov models were jointly used by then for recognition purpose. Jansi et al in [8] proposed compressive sensing-based classification approved for activity recognition. Data from tri-axial accelerometer sensor was initially compressed by them before classification. In [9], classification was done based on sparse theory. Data from different were used and utilized for the classification. Authors have taken into account both the time domain and frequency domain features for training the classifiers. Li et al in [10], proposed a recognition algorithm to classify both basic and transitional activities. In [11], authors have used skeletal data provided by the depth sensor for classification. Most popular deep neural network ResNet was used by the for classifications and they tested their algorithm on available benchmark data sets. A survey on assisted living tools for older adults was presented by authors in [12]. Their proposed work struggles in recognizing similar gestures.

In this paper, skeletal data-based activity recognition system is presented. Skeletal data of kinect sensor is used for classification purpose. Set of statistical features are computed from the skeletal data. The dimensions of statistical features computed are reduced using Principal component analysis (PCA) approach. PCA is a technique used for finding smaller number of uncorrelated data is employed for reducing the feature dimension. The dimensionally reduced features are used as training data for training the classifier. The ability of K-nearest neighbor (KNN) classifier and Support vector machine (SVM) classifier in classifying actions are analyzed. The proposed classification method is tested on most popular KARD (Kinect activity recognition dataset) dataset and various classification parameters are computed. The paper is structured as follows: Section II details the proposed work. Experimental results and analysis are discussed in section III and conclusion is given in section IV.

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II. Classification Method

Fig. 1 shows the block diagram of the presented classification method. Kinect's skeletal data are used for classification purpose. Eight statistical features are extracted from the skeletal data sequence. The dimensions of these features are reduced using principal component analysis. The dimensionally reduced feature are given to classifier for training purpose and same set of features are used for testing and recognizing actions. In our work, we have used kinect activity recognition dataset (KARD) [13] for classification purpose. The dataset consists of 18 actions performed 3 times by 10 different subjects. Thus, for each action there are 30 data. As suggested by the authors of the dataset, the whole database is split into three different group; Group 1, Group 2 and Group 3with each set consists of eight actions. Table I shows the Group of activities in each dataset

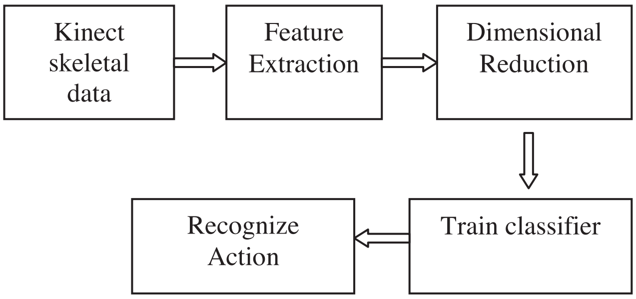

Fig. 1. Block diagram of classification method

TABLE I
KARD DATASET

S.no	Group 1	Group 2	Group 3
1	Horizontal arm wave	High-arm wave	Draw tick
2	Two hand waves	Side kick	Drink
3	Bend	Catch up	Sit down
4	Phone call	Draw tick	Phone call
5	Stand up	Hand clap	Take umbrella
6	Forward Kick	Forward Kick	Toss paper
7	Draw X	Bend	High throw
8	Walk	Sit down	Horizontal arm wave

A Feature Extraction

Following eight statistical features are considered for training the classifier.

1. Simple mean (μ) given by ratio of sum of all values to total no. of values.

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i \quad (1)$$

2. Root mean square (*RMS*) is defined as the total sum of square root of each data in an observation.

$$RMS = \sqrt{\frac{\sum_{i=1}^{n} x_i^2}{n}} \quad (2)$$

3. Standard deviation (σ) is defined as the variation of the values or data from an average mean. It is given by the equation 3

$$\sigma = \sqrt{\frac{1}{n}}(x_i - \mu)^2 \quad (3)$$

4. Index of dispersion (*D*) is usually computed as ratio of the variance to the mean. It is given by the equation 4

$$D = \frac{\sigma^2}{\mu} \quad (4)$$

5. The absolute (*A*) difference between two variables

$$A = |x - y| \geq 0 \quad (5)$$

6. The maximum magnitude (*P*) of the deflection of a wave is called amplitude

$$P = A \times \sin\,(2 \times \pi \times f \times t) \quad (6)$$

7. A Skewness (*S*) is a estimate of the asymmetry of the probability distribution of a data about its mean.

$$S = \frac{\sum_{i=1}^{n}(x_i - \mu)^3/n}{\sigma^3} \quad (7)$$

8. Kurtosis (*KT*) is an estimate used to find out extreme values in the tails of a distribution compare to normal distribution.

$$KT = \frac{\sum_{i=1}^{n}(x_i - \mu)^4/n}{\sigma^4} \quad (8)$$

B Dimensional Reduction

The dimensions of the features are reduced to enhance the performance of the classifier. Principal Component Analysis, (PCA) is used for this purpose. PCA is a method for finding smaller number of uncorrelated data from a huge set of given data. These uncorrelated data are usually termed s principal components PCA is thus considered as a powerful tool to emphasize variations in the given data set. It is widely used in variety of applications such as in computer vision and image processing, recognition tasks etc. The steps is involved in PCA as follows.

Step 1: Find the mean vector.

Step 2: Assemble all the data samples in a mean adjusted matrix.

Step 3: Create the covariance matrix.

Step 4: Compute the Eigen vectors and Eigen values.

Step 5: Sort eigen values and their corresponding eigen vectors in descending order.

Step 6: Select top 'k' eigen values

Step 7: Eigen vectors to top 'k' eigen values forms the reduced feature map.

C Train and Classifier

Since the proposed activity recognition work involves very less data for classification (thirty data peraction) conventional SVM and KNN classifiers were used to recognize action compared to popular deep learning techniques. KNN classifier uses K-means Clustering algorithm that determine the number of points taken into consideration for classification. The algorithm involves computation eucledian distance between the data points and selects K points for classification. The choice of K completely depends upon the data. SVM is defined by a separating hyperplane and it is a form of a discriminative classifiers. In recent years, SVM has emerged out as a successful classifier in classifying problems involving very large dataset.

III. EXPERIMENTAL RESULTS AND DISCUSSION

The ability of two popular classifiers namely KNN classifier and SVM classifier in classifying actions with dimensionally reduced features is evaluated. Confusion matrix of the presented classification scheme on three action sets of KARD dataset are computed separately. Confusion matrix of the presented classification scheme on three actions sets of KARD dataset are computed separately. A confusion matrix is a most popular means of describing a classification model and a classifier provided by the confusion matrix aids in easy identification and visualization of confusion between classes eg.one class is commonly mislabelled as the other. Based on the confusion matrix entries, various classification parameters namely accuracy, precision, sensitivity, specificity, F-score are computed**.**

- Accuracy defines how close the measured value is to a known value.

$$ACC = \frac{TP+TN}{TP+TN+FP+FN} \tag{9}$$

- Precision computes the separately of measurements under unchanged condition.

$$PPV = \frac{TP}{TP+FP} \tag{10}$$

- Sensitivity evaluates the ability of the classifier in correctly identifying a true case.

$$TPR = \frac{TP}{TP+FN} \tag{11}$$

- Specificity is a measure the proportion of actual negatives that are correctly identified.

$$TNR = \frac{TN}{TN+FP} \tag{12}$$

- F-score is computed as the harmonic mean of precision and recall.

$$F - scrore = \frac{2*PPV*TPR}{PPV+TPR} \tag{13}$$

TP, TN, FP, FN are True Positive, True Negative, False Positive, False Negative respectively.

Table II shows the confusion matrix entries and classification parameters, precision and sensitivity computed for Group 1 actions with KNN classifier. It can be inferred from the table that around 95 % sensitivity and 100% precision is obtained for action 3 and action 4 of Group 1 action. The other metric parameters namely accuracy, specificity and F-score computed are given as bar plots in Fig. 2 It can be inferred from the bar plots that except for action 6, the accuracy obtained with KNN classifier is above 95 % for all the actions. F-score and specificity is also greater than or equal to 95 % for all the actions except for action 3.

Table III shows the confusion matrix entries and the classification parameter namely the sensitivity and precision for Action set 1 with SVM classifier. Fig. 3 shows the bar plot of other parameters computed for action set 1 with SVM classifier.

TABLE II
CONFUSION MATRIX OF GROUP 1 ACTIONS OBTAINED USING KNN CLASSIFIER

Actions	Act1	Act2	Act3	Act4	Act5	Act6	Act7	Act8	Accuracy in (%)
Act1	26	2	0	0	0	0	1	1	91
Act2	3	26	0	0	0	0	0	1	89
Act3	0	0	30	0	0	0	0	0	100
Act4	0	0	0	30	0	0	0	0	100
Act5	0	0	0	0	27	3	0	0	95
Act6	0	1	0	0	0	29	0	0	98
Act7	4	0	0	0	0	0	26	0	91
Act8	2	0	0	0	0	0	0	28	94
Precision in (%)	91	89	95	80	94	98	98	96	94.25

TABLE III
CONFUSION MATRIX OF GROUP 1 ACTIONS OBTAINED USING SVM CLASSIFIER

Performance measures	KNN Classifier			SVM Classifier			Average	
	Group 1	Group 2	Group 3	Group 1	Group 2	Group 3	KNN	SVM
Accuracy in (%)	92	95	97	99	98	97	94.5	98
Precision in (%)	90	86	89	95	99	97	89	97
Specificity in (%)	87	80	82	85	87	89	84	87
Sensitivity in (%)	75	69	70	75	76	79	70	76
F-score in (%)	80	78	86	81	83	87	80	85

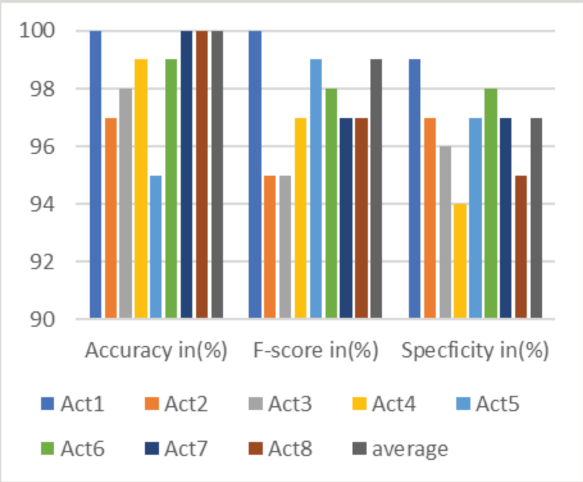

Fig. 2. Accuracy, F-score and Specificity of KARD datasets obtained with KNN classifier for Group1 actions

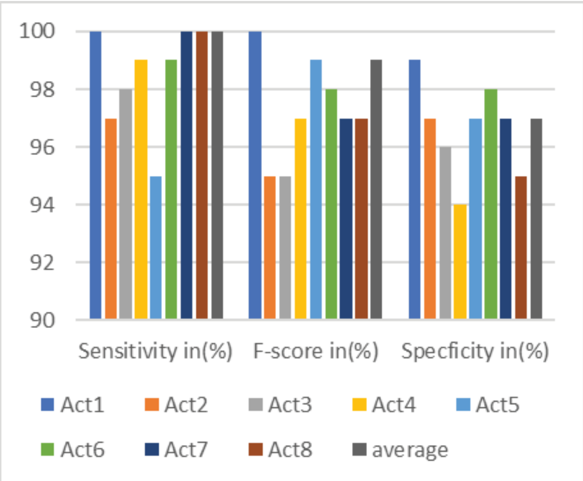

Fig. 3. Sensitivity, F-score and Specificity of KARD datasets obtained with SVM classifier for Group1

TABLE IV
Comparsion Of Classification Parameters

Actions	Act1	Act2	Act3	Act4	Act5	Act6	Act7	Act8	Sensitivity in (%)
Act1	26	0	4	0	0	0	0	0	91
Actt2	0	30	0	0	0	0	0	0	89
Act3	3	0	26	0	1	0	0	0	95
Act4	2	0	2	26	0	0	0	0	95
Act5	0	0	0	0	29	0	0	1	91
Act6	0	1	0	0	0	29	0	0	92
Act7	0	0	0	0	0	0	30	0	91
Act8	0	0	0	0	0	0	0	30	94
Precision in (%)	94	92	95	97	94	99	94	98	95

It can be inferred that sensitivity is above 95 % for all the actions considered in action set 1 and is 100 % for four individual actions. 100% accuracy is obtained for two actions. F-score is above 95 % for all the actions. Except for two actions, specificity of 95 % and above is achieved. Table IV compares the average classification metrics obtained with KNN classifier and SVM classifier for all the 3 action sets. Specificity, Sensitivity, F-score and precision obtained with SVM classifier is 3%, 6%, 5% and 8% higher than the values obtained with KNN classifier. Among the two classifiers considered, SVM classifier performed better with an average overall accuracy of 98% which is 3.5% higher than the accuracy produced by KNN classifier.

IV. CONCULSION

Human activity recognition method with kinect skeletal data is presented. Eight statistical features are extracted from the raw skeletal data for classification purpose. The dimensions of the extracted features are reduced using PCA method. The popular benchmark dataset namely the Kinect activity recognition dataset is used for testing the classifier in recognizing actions.The performance of two classifiers, KNN and SVM in recognizing actions are compared. Among the two classifiers, SVM performed better with an overall accuracy of 98 %.

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